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Material Classification based on Non-contact Ultrasonic Echo Signal Using Deep Learning Approach

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Abstract

Non-contact ultrasonic sensors are generally used for range measurement and object detection. In addition to the shape and size of the target object, the identification of the material types plays a vital role in robotic navigation and autonomous vehicle applications. Ultrasonic echo signals contain a significant amount of information and can be used to recognize and categorize different materials. Echo signals can help robots detect the obstacle's material on the path and comply with its behavior accordingly. The emergence of deep neural networks has shown great promise, offering cutting-edge performance for a wide range of signal-processing tasks. This work uses the non-contact ultrasonic echo signal from a set of materials (glass, wood, metal plate, sponge, and cloth) for classification. The main idea is to classify the materials using the embedded information of reflected echo signals. Hilbert transform is used to get the envelope from the raw echo ultrasonic signal. In this paper, a novel architecture for one-dimensional convolutional neural network (1D-CNN) has been proposed to accomplish the classification task. The CNN model takes raw echo signals as input to detect and classify materials accurately. The performance of the classifier model is evaluated using accuracy, precision, recall, and F1-score. The proposed 1D-CNN deep learning-based multi-class classifier model can classify the different types of materials with an accuracy of 96%, precision of 95%, recall of 95%, and F1-score of 95%.

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Keywords: Ultrasonic echo signal; Material classification; Envelope signal; Hilbert transform; Convolutional Neural Networks (CNNs).

1. Introduction

Ultrasonic sensors are widely used for detection tasks and environment perception [1]. These sensors are also used to solve the problems of autonomous mobile robot navigation, mapping, detection, manufacturing, automotive, and localization tasks [2, 7]. Ultrasonic sensors offer distinct advantages over other sensing techniques due to their cost-effectiveness, ease of implementation, and non-contact nature. Environmental temperature and humidity are the two parameters that influence the sensor's sensing performance [3]. The most common use of an ultrasonic sensor is to measure the Time of Flight (ToF) based on the time it takes for an ultrasonic pulse to travel from the sensor to the target object and back the reflected echo signal. The reflected echo amplitude signal contains significant information about the target object or surface. The distance and the strength of the reflected echo ultrasonic signal play an important role

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in accurately characterizing materials, which helps the robot move seamlessly. Therefore, highly accurate material identification is essential for mobile robot navigation. Classification using conventional machine learning techniques needed multiple sequential steps: feature extraction, feature selection, model learning, and classification. Moreover, feature selection mostly affects machine learning techniques' performance. Auto feature extraction is an inherent characteristic of the deep neural network models compared to other conventional machine learning techniques [8, 9]. This work uses the raw echo envelop signal to input a 1-D CNN model without needing an extra feature extraction module [4, 6].

A single ultrasonic sensor can classify four targets (e.g., edge, plan, small cylinder, and corner) in indoor environments based on echo signal analysis [10]. A multi-transducer pulse-echo ranging system differentiates between a plane or a right angle corner [11]. An array of ultrasonic sensors with an artificial neural network is used to detect and classify objects with different shapes, such as circular, triangular, and square shapes [12]. Classification of ten different materials using different classification algorithms based on capacitive proximity sensor signals for robotic navigation is proposed in [13]. A fuzzy ARTMAP neural network classification system is used to recognize objects at varying distances using ultrasonic echo signal [14]. The raw ultrasonic signal is used to classify objects by implementing different machine learning algorithms [15]. A novel method for classifying objects and recognizing the properties of materials, such as texture and density information, by considering the reflected echo signal is proposed in [16]. Echo envelope signals are used to locate the object and identify the object using the features of the echo signal [17]. An algorithm for distance measurements and object identification using the reflected echo of the ultrasonic transducers is proposed in [18]. In [19], authors have proposed a neural network approach to recognize the object using the features extracted from the echo signal. A combination of an array of ultrasonic sensors and a neural network algorithm to identify the shape and size of an object in a non-contact manner is proposed in [20]. In [21], authors have proposed ultrasonic-based detection of the orientation and size of objects for robotic applications. Ultrasonic sensor combined with a multi-layer perception model is also used to classify different objects [22]. Echo signal feature information used to identify objects helps to develop ultrasonic mobility aid to assist blind people [23].

In this work, a new 1-D CNN model is proposed to classify five different materials, such as glass, steel, wood, cloth, and sponge, using the raw ultrasonic echo signal. This work aims to distinguish materials based on the echo signal amplitude of ultrasonic sensors with potential applications to robotic navigation and automotive applications. The novelty of this research is the utilization of one-dimensional CNN configurations to classify the materials with higher accuracy efficiently. To the best of our knowledge, this is the first kind of work that used deep learning models to ultrasonic echo signals for material classification tasks. After reviewing the existing literature certain research gaps have been identified that are worth further investigation. The major research gaps are the following.

- Most works focus on the classification and recognition of the shape of target objects but not the material types.
- A few research focuses on material classification using other types of sensors, which are not feasible to work in all environments.
- Some of the existing work used ultrasonic signals for classification based on a complex signal processing approach.
- Most machine learning algorithms are used for classification, which requires feature extraction, and the system's performance depends largely on the type of features selected.

Thus, the main objective of the work is to design and develop a deep-learning model for material classification using raw echo signals of a low-cost ultrasonic sensor.

The major contributions of this research are the following:

- Designed and developed an experiment to capture the ultrasonic echo signals and determine the envelope signal using the Hilbert Transform model.
- Conducted exhaustive experiments and collected signals from a set of materials like glass, metal, cloth, sponge, and wood.
- Proposed a 1-D CNN model architecture for efficient classification of different materials.

The rest of the paper is organized as follows. Section II describes the overview of ultrasonic time domain signal processing, envelope extraction using the Hilbert transform, and the basics of the CNN model. The proposed method and the CNN configuration parameters are described in Section III. Section IV describes the experimental setup and procedure for data collection and reprocessing. Section V presents the experiment's results to explain the proposed method's effectiveness. Finally, Section VI summarizes and concludes the work with future research scope.

2. Background

Ultrasonic measurements are based on calculating the Time of Flight (ToF). In addition to range measurement, echo signals contain significant information that can be used for object identification and material classification. The echo waveform ultrasonic signal in the time domain can be converted into frequency using the Fast Fourier Transform (FFT). The raw reflected echo signal of the ultrasonic sensor can be processed using time and frequency domain analysis as shown in Figure 1.

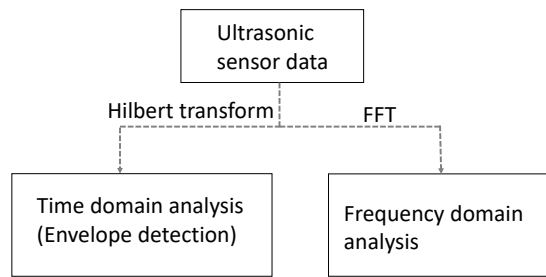


Fig. 1: Ultrasonic signal processing steps

The time-domain signal is the raw data form of an ultrasonic sensor. The envelope contained in the raw ultrasonic signal can be extracted using Hilbert transform [24, 25]. In the time domain, the signal's envelope represents the signal's amplitude. An envelope is the extreme peak point in the waveform within which the signal is contained. The raw signal fed into the Hilbert transforms to generate an analytical signal, which helps to determine the signal's envelope as shown in Figure 2. The standard procedure to calculate the envelope from a raw input signal is described in Algorithm 1. In the time domain, the amplitude of the received echo signal is an envelope that starts from the baseline, reaches a peak, and then goes back to the baseline of the signal.

Algorithm 1 Classical approach for envelope detection

Input : Raw echoSignal

Output : Envelope

```

1: env_array ← [ ]
2: baseline ← average(echoSignal)
3: for index, value in echoSignal do
4:   if value > baseline then
5:     env_array.push(index)
6:   end if
7: end for
8: return env_array
  
```

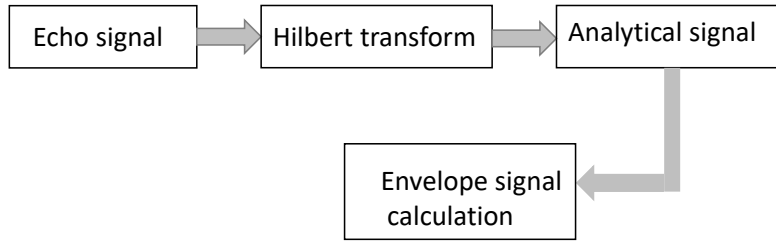


Fig. 2: Steps involved in the calculation of envelope signal using Hilbert transform

2.1. Hilbert Transform (HT)

The Hilbert transform technique analyzes various non-stationary signal and image processing applications. An analytical signal is the complex-valued representation of a real-valued signal. The essential role of the Hilbert transform is to filter out the negative frequency components from the raw input signal without disturbing the phase value. The applications of HT includes biomedical, ultrasonic, radar, and speech recognition system.

The Hilbert transform is defined as follows if $y(t)$ is an original echo signal:

$$h(t) = H\{y(t)\} = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{y(\tau)}{t - \tau} d\tau = y(t) * \frac{1}{\pi t} \quad (1)$$

The input signal $y(t)$ serves as the real component of the input signal, while its Hilbert transform, $h(t)$, serves as the imaginary part, as shown in the following:

$$y_a(t) = y(t) + jh(t) \quad (2)$$

where, $y_a(t)$ is the analytical signal.

The envelope $E(t)$ of the input signal is calculated as follows:

$$E(t) = |y(t) + jh(t)| = \sqrt{y^2(t) + h^2(t)} \quad (3)$$

$$phase(\Phi) = \arctan\left(\frac{h(t)}{y(t)}\right) \quad (4)$$

This work attempts to differentiate five different materials using ultrasonic echo signals. Hilbert transform calculates the amplitude envelope of the raw ultrasonic signal. Figure 3 depicts the ultrasonic echo envelope estimation using Hilbert transform.

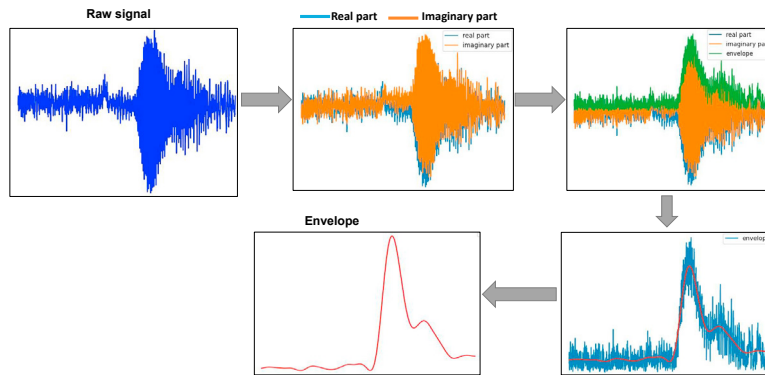


Fig. 3: The process of extraction of an envelope from the raw signal using Hilbert transform

2.2. Convolutional Neural Network (CNN)

Convolutional neural networks are a type of artificial neural network specifically designed for image classification tasks [9]. CNNs comprise multiple layers of interconnected neurons, with each layer responsible for extracting a specific set of features from the input data. The layers at the beginning of the network typically consist of convolutional layers, which apply a set of filters to the input data to detect various features, such as edges, corners, and patterns. These features are then passed through pooling layers, which reduce the feature map's size and the network's computational complexity. The final layers of a CNN are typically fully connected, combining the features extracted by the convolutional and pooling layers to make a final prediction or classification. CNNs are effective at image classification tasks because they can automatically learn and extract the relevant features from the input data rather than requiring manual feature extraction by the user. While CNN remains the most commonly used deep learning architecture, the development of problem-specific CNN architectures is still increasing.

In addition to image classification, CNNs are used in other applications such as object detection, signal classification, semantic segmentation, and natural language processing. They are widely used in various fields due to their ability to automatically learn useful features from data and their effectiveness at solving complex tasks. This study introduces a groundbreaking 1-D CNN model for the classification of materials, leveraging the echo envelope signal obtained from a non-contact ultrasonic sensor.

3. Methodology

This section describes the proposed method along with a comprehensive analysis of the CNN layers employed in the proposed architecture. The overall system flow of this work is shown in Figure 4. A one-dimensional CNN model is proposed to classify materials using ultrasonic echo signals. Figure 5 shows the architecture of the one-dimensional convolutional neural network.

The convolution layer is regarded as the fundamental building block of CNN architecture. It plays a pivotal role in the detection of diverse features by applying a set of filters to the input data [5]. Convolutional layers are typically followed by the non-linear activation function ReLU (Rectified Linear Unit) function, which introduces non-linearity to the network and allows it to learn more complex features. A pooling layer reduces the size of the feature map produced by the convolutional layers by down-sampling the data. Pooling is typically applied after a convolutional layer to reduce the computational complexity of the network. ReLU and softmax functions are used as activation functions. ReLU is used after each convolution layer, and The classifier networks use the softmax function for the final output. The dropout layer randomly deactivates certain neurons in the network, rendering them inactive during the learning process, thereby not impacting the network's training. The fully connected layer consists of many interconnected neu-

rons. The process involves taking the output of the preceding layers, flattening them, and converting them into a single vector that can serve as an input for the subsequent phase.

Each input signal size is 1999, with a total of 730 signal frames collected for training and testing. Five class labels (class-1: cloth, class-2: glass, class-3: metal plate, class-4: sponge, class-5: wood) are used in this model, each corresponding to different material types. The entire dataset is divided into two different datasets with sizes of 600 data for training and 130 data for testing. The one-hot encoder encodes each data point before sending it to the model. The first two layers have 16 features of an eight kernel size and a stride of 2 with the same padding. Then a max pool layer of size two is followed by two convolution layers of 64 features of each kernel size four and a stride of 2 with the same padding. This is followed by a max-pooling layer of size 2, the same as the previous layer. Then there are two convolution layers of kernel size 4 and 256 features and a max pool layer of size 2. After that, there are two convolution layers and a max pool layer. Each has 512 features of 2 kernel sizes with the same padding and a stride of 2. In all the convolution layers, ReLu is used as an activation function.

After the stack of convolution and max-pooling layer, There is a Global-Average-Pooling of size 512 and dropout of 0.3. After that, there is the dense layer (fully connected) of size 5 with the activation function SoftMax.

Here, the loss function is categorical cross-entropy, and the optimizer employed is Adam. The batch size is set at 20 and the number of epochs is also set at 20. The proposed 1-D CNN structure is shown in Figure 6. Detailed parameters for training and validation of the model are described in Table 1.

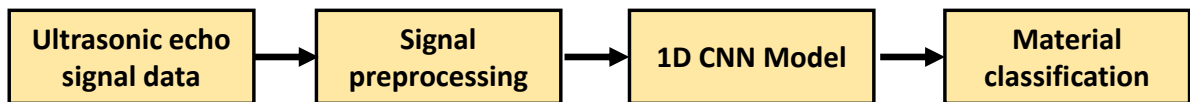


Fig. 4: System flow of material classification using proposed approach

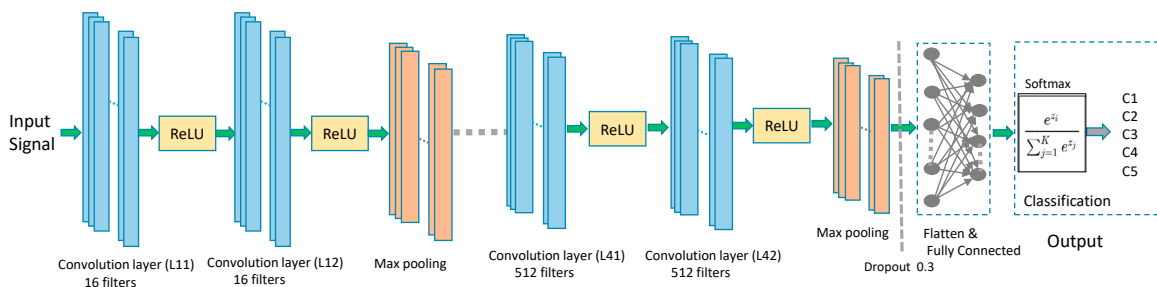


Fig. 5: 1-Dimensional convolutional neural network architecture

Table 1: Model configuration parameters and the corresponding values

Layers	Configurations	Output Dimensions
Convolution (C11)	16 Filter, 8 Kernel, ReLu	(None, 996, 16)
Convolution (C12)	16 Filter, 8 Kernel, ReLu	(None, 498, 16)
Max Pooling (p1)	2 kernel	(None, 249, 16)
Convolution (C21)	64 Filter, 4 Kernel, ReLu	(None, 125, 64)
Convolution (C22)	64 Filter, 4 Kernel, ReLu	(None, 63, 64)
Max Pooling (p2)	2 kernel	(None, 31, 64)
Convolution (C31)	256 Filter, 4 Kernel, ReLu	(None, 16, 256)
Convolution (C32)	256 Filter, 4 Kernel, ReLu	(None, 8, 256)
Max Pooling (p3)	2 kernel	(None, 4, 256)
Convolution (C41)	512 Filter, 2 Kernel, ReLu	(None, 4, 512)
Convolution (C42)	512 Filter, 2 Kernel, ReLu	(None, 4, 512)
Max Pooling (p4)	2 kernel	(None, 2, 512)
Global Average Pooling	–	(None, 512)
Dropout (D)	0.3	(None, 512)
Dense Layer (FC)	SoftMax	(None, 5)

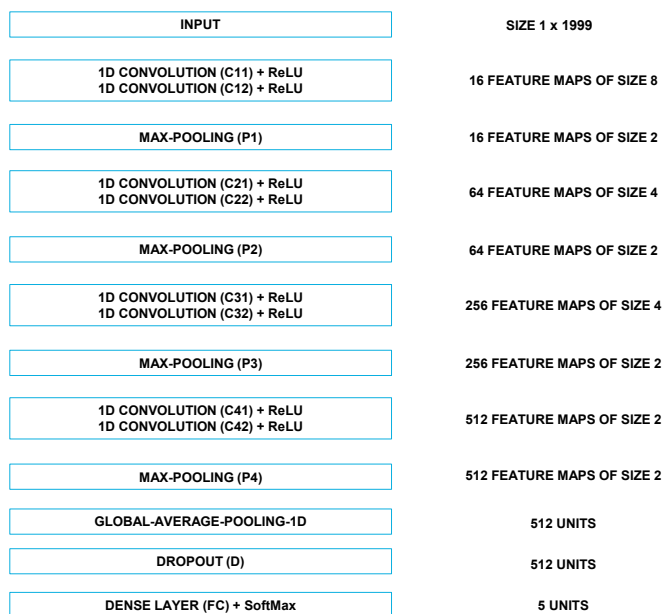


Fig. 6: Structure of the proposed 1-D CNN model

4. Experimental Setup and Data Collection

A 40 kHz ultrasonic sensor (HC-SR04) is used for this experiment. The experimental setup consists of an ultrasonic sensor, a microcontroller, and an oscilloscope. The setup diagram is shown in Figure 7. The oscilloscope is used to investigate the acquired sensor echo signal more deeply. Moreover, the oscilloscope displays and digitizes the interference signals, which are finally stored on the computer. The sampling rate at which the signal is sampled is 200 MSamples/second, which is higher than the Nyquist rate of 80 KSamples/second to avoid loss of information.

Here, five materials: glass, metal plate, wood, cloth, and sponge are used for the experimentation. The reflected echo signal of the ultrasonic sensor is acquired for all five materials at different distances. The reflected raw echo signal of the ultrasonic sensor is captured and preprocessed before being provided as input to the proposed 1D-CNN model for classification. The preprocessing steps involve DC-offset removal and noise filtering. To remove the DC offset from the signal is a matter of subtracting the mean. For a one-dimensional signal, it is expressed as

$$\text{balanced signal} = \text{original signal} - \text{mean}(\text{original signal}) \quad (5)$$

The expression for a one-dimensional signal model with white noise can be represented as follows:

$$s(k) = f(k) + \varepsilon(k) \quad (6)$$

Where $s(k)$ represents the observed signal with noise component, $f(k)$ is the required signal for classification, and $\varepsilon(k)$ denotes the white Gaussian noise component follows normal distribution $N(0, \sigma^2)$. This $\varepsilon(k)$ is generally a high-frequency signal component, and $f(k)$ is the required stable signal with a low-frequency component. Therefore to remove this high-frequency noise, a low pass filter is used to filter out the noise. A scenario of the experiment to capture the echo signal of the material under test using an oscilloscope is shown in Figure 8. The experiment is performed with different types of materials 600 times and labeled with class labels ranging from 0 to 4 to represent different types of materials. The dataset is almost balanced with an equal number of sample points for each type of material.

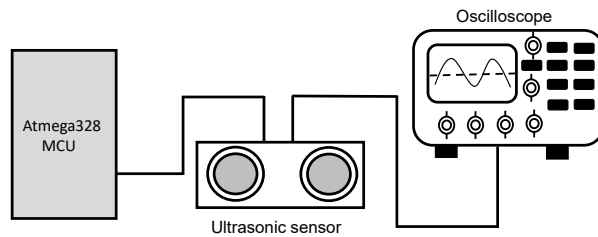


Fig. 7: Experimental setup for extraction of echo envelope signal from ultrasonic sensor

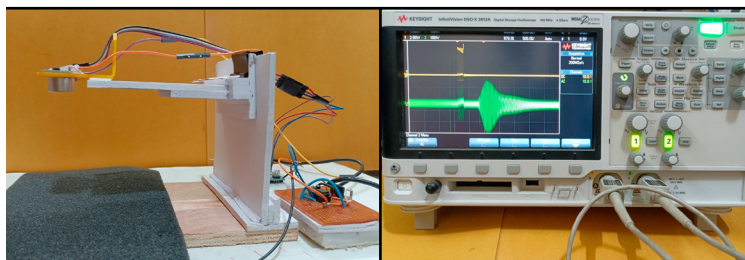


Fig. 8: Experimental setup to capture the echo signal for different materials under test using oscilloscope

5. Results and Discussions

This experiment considers five different materials: cloth, glass, metal plate, sponge, and wood. These materials' reflected ultrasonic echo signal is recorded and fed into the Hilbert transform. The DC offset is eliminated from the signal prior to the application of the Hilbert transform. The envelopes extracted using the Hilbert transform are shown in Figure 9. It is observed from the envelop signals the materials are distinguishable.

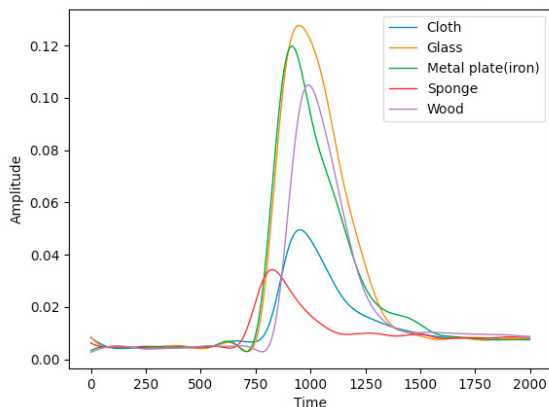


Fig. 9: Envelope signals for five different classes of materials, where the x-axis is the time in *milliseconds* and the y-axis is the *amplitude*

The proposed 1-D CNN model is trained and validated with 600 data points. The dataset is partitioned into an 80:20 ratio for training and validation. The validation is performed to fine-tune the hyper-parameters of the proposed model. For the training process, 150 epochs were primarily utilized, although most architectures showed convergence between 50 and 100 epochs. To ensure the architecture's consistent performance on both training and test sets, 50 repetitions were conducted for each evaluation. Table 1 shows the configuration parameters of the best model. The model's performance achieves 98% for training and 95% for validation accuracy, with 9% training loss and 32% validation loss as shown in Figure 10. Table 2 shows the dataset size training, validation, and testing corresponding accuracies. The training and validation loss indicates how the data fit the model over the number of epochs. Figure 10 (a) shows the progression loss during the training phase. Figure 10 (b) shows how the validation accuracy approaches the intended accuracy after a few epochs.

Table 2: Performance accuracy of Training, Validation and Testing

	Size	Accuracy
Training	480	98%
Validation	120	95%
Testing	130	96%

Here, 130 previously unobserved data points are utilized for testing. In most cases, the proposed model can successfully classify these testing data frames, and out of 130 testing data frames, the system can classify 126 frames accurately. The confusion matrix is generated utilizing the test data samples. From the confusion matrix, the performance accuracy is calculated, which is 96%. Figure 11 shows the confusion matrix. The number of test instances in each class is 23 (Cloth), 21 (Glass), 41 (Metal), 23 (Sponge), and 22 (Wood), respectively. The confusion matrix shows very few classification errors. It is observed that in very few cases, the test data samples are misclassified. This is due to the similar envelope structure of the material class-1 (cloth) and class-4 (sponge). Similarly, for material class-2 (glass) and class-3 (metal plate). The accuracy, precision, recall and F1-score of the proposed 1D CNN model are presented in Table 3.

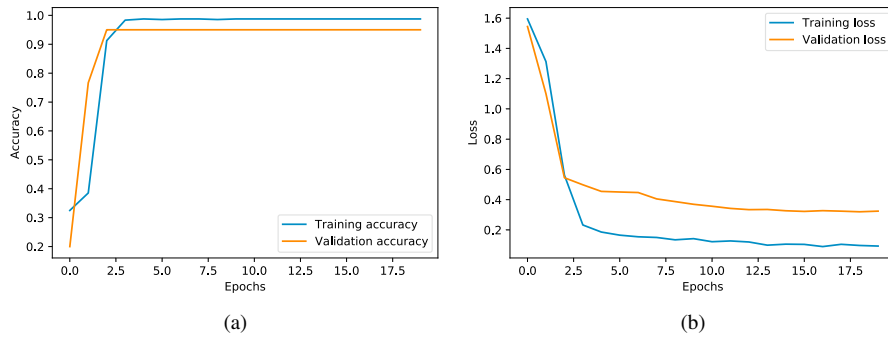


Fig. 10: (a) Training loss and validation loss of the convolutional neural network versus the number of epochs, (b) Training and validation accuracy of CNN versus the number of epochs

Table 3: Performance metrics (Accuracy, Precision, Recall and F1-score) of proposed 1D CNN model

Algorithm	Accuracy	Precision	Recall	F1-score
1-D CNN	96%	95%	95%	95%

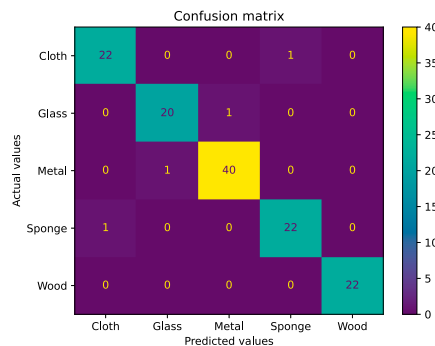


Fig. 11: Confusion matrix for 1D-CNN based material classification using ultrasonic echo envelope signal

6. Conclusions and future works

The purpose of the work is to accurately classify the materials using raw ultrasonic echo signals for autonomous vehicles and robotic navigation applications. The experimental results show that ultrasonic sensors, in combination with the CNN model, have a great potential to classify and recognize different materials and objects. In this experiment, five different materials are considered (glass, wood, metal plate, sponge, and cloth) for classification. More samples per measurement may tend to produce better results. The advantage of this work is the use of 1D CNN, which automatically extracts features from the input data, and a low-cost ultrasonic sensor to accurately classify the materials. The disadvantage of this method is sometimes material like glass is misclassified as a metal plate and vice versa. Similarly, the material cloth is misclassified as a sponge and vice versa. This misclassification is due to the similar structure of envelopes. In addition, ultrasonic sensors have poor response in the presence of electronic noise and environmental attenuation.

In future works, echo signals from various other materials will be considered to train the proposed CNN model to make a more robust classification model. Also, to investigate the frequency and time domain signals using the proposed

method for gesture recognition, object shape determination, object identification, and various material classification tasks. Furthermore, it to compare other convolutional neural network architectures such as GoogLeNet, AlexNet, and ResNet with the proposed method. The combination of ultrasonic sensor data and the camera can be fused to detect, recognize, and classify materials in complex scenarios. Integration of ultrasonic and WiFi sensing techniques can be a promising application for IoT-based smart home solutions.

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References

- [1] A. K. Sahoo and S. K. Udgata, "A Novel ANN-Based Adaptive Ultrasonic Measurement System for Accurate Water Level Monitoring," in *IEEE Transactions on Instrumentation and Measurement*, vol. 69, no. 6, pp. 3359-3369, June 2020, doi: 10.1109/TIM.2019.2939932.
- [2] Qiu, Zurong, Yaohuan Lu, and Zhen Qiu. (2022) "Review of ultrasonic ranging methods and their current challenges", *Micromachines* 13, No. 4: 520. <https://doi.org/10.3390/mi13040520>.
- [3] Ajit Kumar Sahoo, Siba K. Udgata, A novel fuzzy inspired machine learning framework for relative humidity estimation using time-of-flight of ultrasonic sensor, *Measurement*, Volume 195, 2022, 111035, ISSN 0263-2241, <https://doi.org/10.1016/j.measurement.2022.111035>.
- [4] Serkan Kiranyaz, Onur Avci, Osama Abdeljaber, Turker Ince, Moncef Gabbouj, and Daniel J. Inman. (2021) "1D convolutional neural networks and applications: A survey", *Mechanical Systems and Signal Processing*, Volume 151, 107398. <https://doi.org/10.1016/j.ymssp.2020.107398>.
- [5] Khan F, Yu X, Yuan Z, and Rehman AU. (2023) "ECG classification using 1-D convolutional deep residual neural network." *PLoS One*, 18(4):e0284791. doi: 10.1371/journal.pone.0284791.
- [6] Roberto Sánchez-Reolid, Francisco López de la Rosa, María T. López, and Antonio Fernández-Caballero. (2023) "One-dimensional convolutional neural networks for low/high arousal classification from electrodermal activity." *Biomedical Signal Processing and Control*, Volume 71, Part B, 103203. <https://doi.org/10.1016/j.bspc.2021.103203>.
- [7] Mizutani, Koichi, Naoto Wakatsuki, and Tadashi Ebihara. (2016) "Introduction of measurement techniques in ultrasonic electronics: Basic principles and recent trends." *Japanese Journal of Applied Physics* 55.7S1: 07KA02. DOI 10.7567/JJAP.55.07KA02.
- [8] Buduma, Nithin, Nikhil Buduma, and Joe Papa. (2022) *Fundamentals of deep learning*. "O'Reilly Media, Inc".
- [9] Kiranyaz, Serkan, Turker Ince, Osama Abdeljaber, Onur Avci, and Moncef Gabbouj. (2019). "1-D convolutional neural networks for signal processing applications." *IEEE ICASSP 2019*, pp. 8360-8364, IEEE.
- [10] Barat, C., and Oufroukh, N. A. (2001) "Classification of indoor environment using only one ultrasonic sensor." *IEEE IMTC 2001*, Vol. 3, pp. 1750-1755, IEEE. doi: 10.1109/IMTC.2001.929501.
- [11] Barshan, B., and Kuc, R. (1990) "Differentiating sonar reflections from corners and planes by employing an intelligent sensor." *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 12 (6), 560-569. doi: 10.1109/34.56192.
- [12] Barua, Sancyo, Anoy Saha, Ainul Anam Shahjamal Khan, and Rubana H. Chowdhury. (2019) "Comparative Study of Object Shape Recognition using Ultrasonic Sensor Arrays with Artificial Neural Network." *2019 ICIET*, pp. 1-6, IEEE. doi: 10.1109/ICIET48527.2019.9290607.
- [13] Y. Ding, H. Kisner, T. Kong and U. Thomas. (2020) "Using Machine Learning for Material Detection with Capacitive Proximity Sensors," *2020 IEEE/RSJ IROS*, Las Vegas, NV, USA, pp. 10424-10429, doi: 10.1109/IROS45743.2020.9341016.
- [14] Ecemis, M. Ihsan, and Paolo Gaudiano. (1999) "Object recognition with ultrasonic sensors." *IEEE International Symposium on CIRA*, pp. 250-255, IEEE. doi: 10.1109/CIRA.1999.810057.
- [15] Hwasser, Ruben. (2020) *Machine learning classification based on ultrasonic analog data*.
- [16] Kang, Lei, Andrew Feeney, and Steve Dixon. (2020) "The high frequency flexural ultrasonic transducer for transmitting and receiving ultrasound in air." *IEEE Sensors Journal*, vol. 20, no. 14, pp. 7653-7660. doi: 10.1109/JSEN.2020.2981667.
- [17] Kuc, Roman. (1997) "Biomimetic sonar locates and recognizes objects." *IEEE Journal of Oceanic Engineering*, vol. 22, no. 4, pp. 616-624. doi: 10.1109/48.650828.
- [18] Lindstedt, Gunnar, and Gustaf Olsson. (1993) "Using ultrasonics for sensing in a robotic environment." *1993 IEEE ICRA*, vol.3, pp. 671-676. doi: 10.1109/ROBOT.1993.291828.
- [19] Llata, J. R., E. G. Sarabia, and J. P. Oria. (2001) "Pattern recognition with ultrasonic sensors: a neural networks evaluation." *Sensor Review*, Vol. 21 No. 1, pp. 45-51. <https://doi.org/10.1108/02602280110365653>.
- [20] Ohtani, K., and M. Baba. (2006) "A simple identification method for object shapes and materials using an ultrasonic sensor array." *2006 IEEE IMTC*, pp. 2138-2143. doi: 10.1109/IMTC.2006.328525.
- [21] Oria, JM Perez, and A. M. G. Gonzalez. (1993) "Object recognition using ultrasonic sensors in robotic applications." *Proceedings of IECON '93-19th Annual Conference of IEEE Industrial Electronics*, pp. 1927-1931 vol.3, doi: 10.1109/IECON.1993.339369.
- [22] P.D. Smith, D.R. Bull & C. Wykes. (1970) "Target classification with artificial neural networks using ultrasonic phased arrays." *WIT Transactions on Information and Communication Technologies* 1. doi: 10.2495/AIENG930551.
- [23] Worth, Patricia Lee. (2011) *Object detection and classification with a CTFM ultrasonic sensor*.
- [24] Ulrich, Timothy. (2006) "Envelope calculation from the Hilbert transform." *Los Alamos Nat. Lab.*, Los Alamos, NM, USA, Tech. Rep.
- [25] Xu, Wansong, and FANYU DU. (2022) "A robust QRS complex detection method based on Shannon energy envelope and Hilbert transform." *Journal of mechanics in medicine and biology* 22.03 : 2240013.